

TEZPUR UNIVERSITY

Mid-Term Test, Autumn 2014

C0 504 Natural Language Processing

Marks : 40

Time : 60 minutes

The figures in the right-hand margin indicate marks for the individual question.

1. How can a corpus help build a computational model of a language? What are raw and annotated corpora? What are some possible types of annotation of a corpus? What is a parallel corpus? 3 + 2 + 2 + 1
2. How do unigram, bigram, trigram and quadrigram differ in their capability to model a language? What are the practical considerations in selecting any one of them for actual use? 4 + 3
3. Consider the following text segment-

Words are the fundamental building block of language. Every human language, spoken, signed or written, is composed of words. Every area of speech and language processing, from speech recognition to machine translation to information retrieval on the Web, requires extensive knowledge about words. Psycholinguistic models of human language processing and models from generative linguistics are also heavily based on lexical knowledge. [61 words]

- (a) What are the unigram probabilities of the words- 3

words, language, block, human, to, are

- (b) What are the bigram probabilities of the following- 4

are the, human language, language processing, speech synthesis

- (c) How would taking into account punctuation marks in n-gram analysis make a difference? 2

4. What is *add-one smoothing*? 3
5. Why is morphological analysis required? What is the importance of morphotactics? What are stored in a lexicon to be used for morphological analysis? 2 + 2 + 2
6. (a) How are the following distinct- Unicode, UTF-16, and “Times New Roman” fonts? 4
- (b) What is sampling of a signal? 3